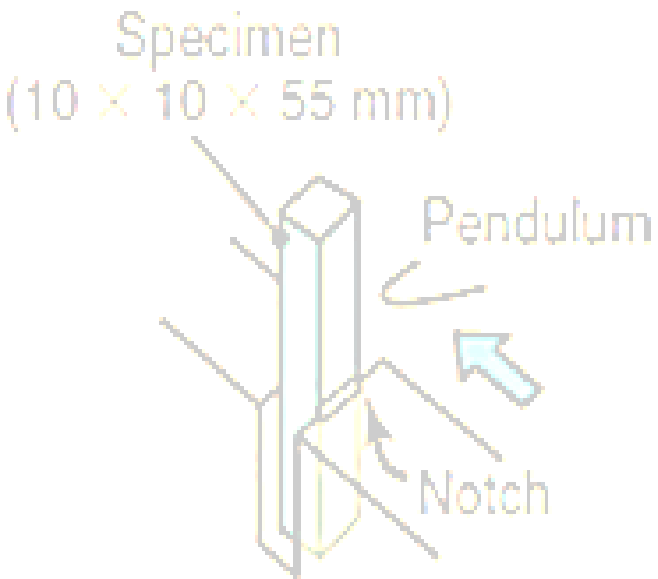


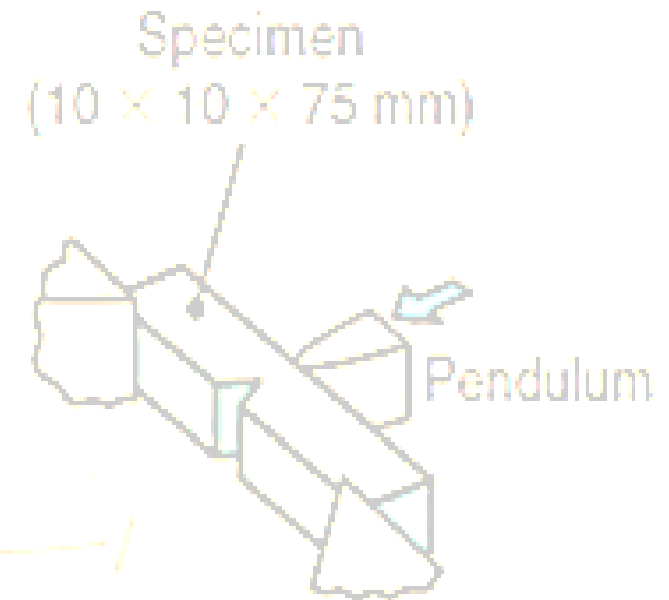
Lab No 09

Impact Testing



Izod

Impact Testing Machine



Charpy

Mechanical Properties of Materials

- Mechanical behavior of materials reflects how they respond to applied loads or forces.
- Important mechanical properties include strength, hardness, ductility, stiffness, toughness, and resilience.
- Strength measures the capacity of an object or substance to withstand great force or pressure.
- Hardness indicates how resistant solid matter is to shape changes under applied force.

Cont...

- Ductility is the measure of material's plastic deformation prior to fracture
- Stiffness refers to the physical property of being inflexible and hard to bend.
- Toughness (also known as impact toughness or fracture toughness) is the ability of a material to absorb energy and deform plastically without fracturing.
- Resilience is the ability of a material to absorb energy when deformed elastically.

Property	Definition	Measured by	Key Relation	Example
Hardness	Resistance to indentation or scratching	Hardness tests (Rockwell, etc.)	Often increases with strength	Glass (hard but brittle)
Strength	Ability to resist breaking under load	Stress at failure (MPa)	↑ Strength → ↓ Ductility (usually)	Steel
Ductility	Ability to deform without breaking	% Elongation or reduction area	Opposite of brittleness	Copper (high ductility)
Stiffness	Resistance to elastic deformation	Young's Modulus (E)	Not same as strength or toughness	Diamond (very stiff)
Toughness	Total energy absorbed before fracture	Area under stress-strain curve	Toughness = Strength + Ductility	Rubber (tough, not hard)
Resilience	Energy absorbed in elastic range (springiness)	Area under elastic region	Part of toughness (only elastic portion)	Spring steel

Problem:

What happened to the great Titanic, back in 1912?

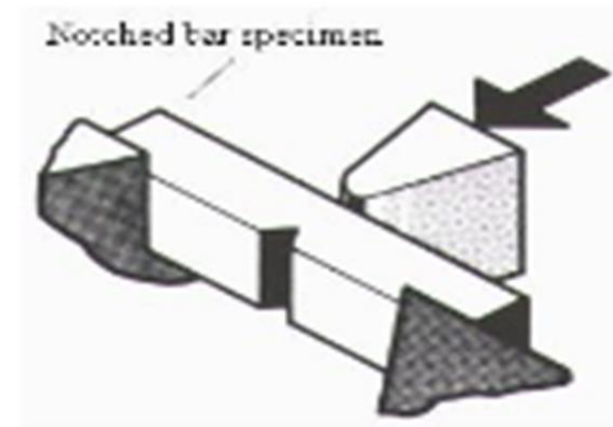
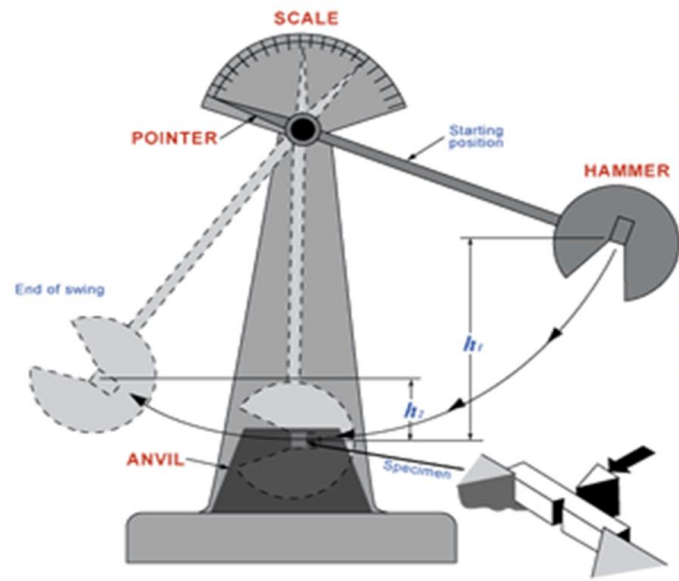
Assumption:

Ductile material of the ship's sole transformed to brittle material due to low temperature

- Ductile to Brittle Transformation (DBT): Materials transforms to brittle from ductile nature over a range of temperatures
- Mild steel exhibits a ductile-brittle transition around $-60\text{ }^{\circ}\text{C}$.

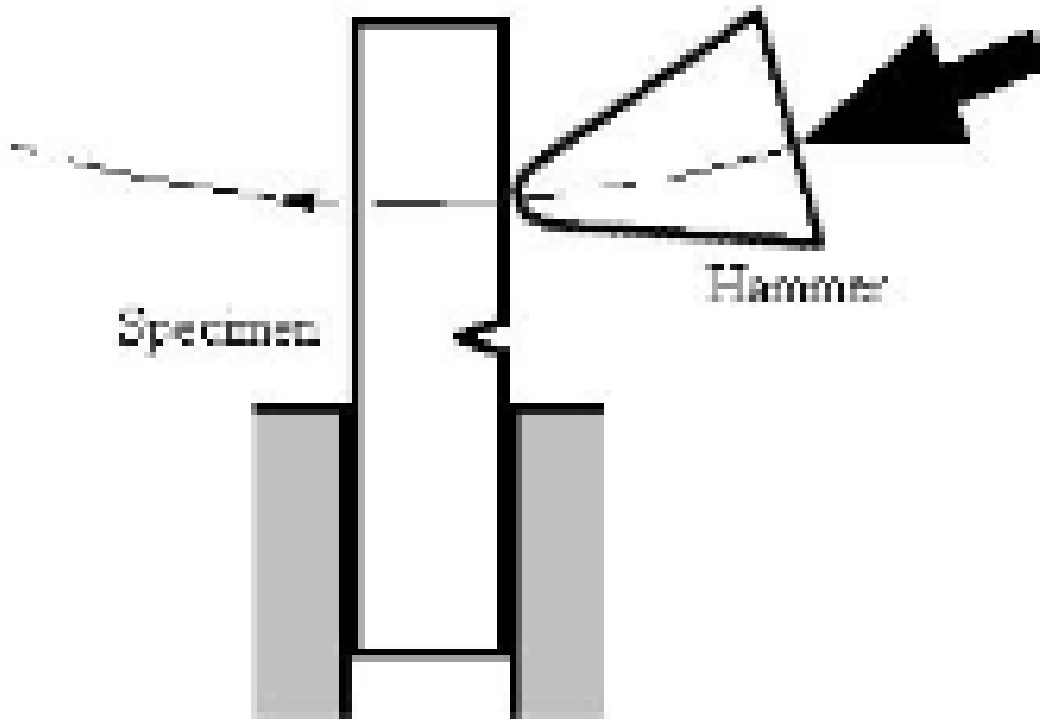


Structure	Ductility	Toughness	Low Temp Behavior	Examples
FCC	High	High	Remains ductile	Aluminum, Copper, Gold, Austenitic Stainless Steel
BCC	Lower	Lower	Becomes brittle (DBTT effect)	Ferritic Steel, Mild Steel, Tungsten, Iron (room temp)



Charpy test configuration

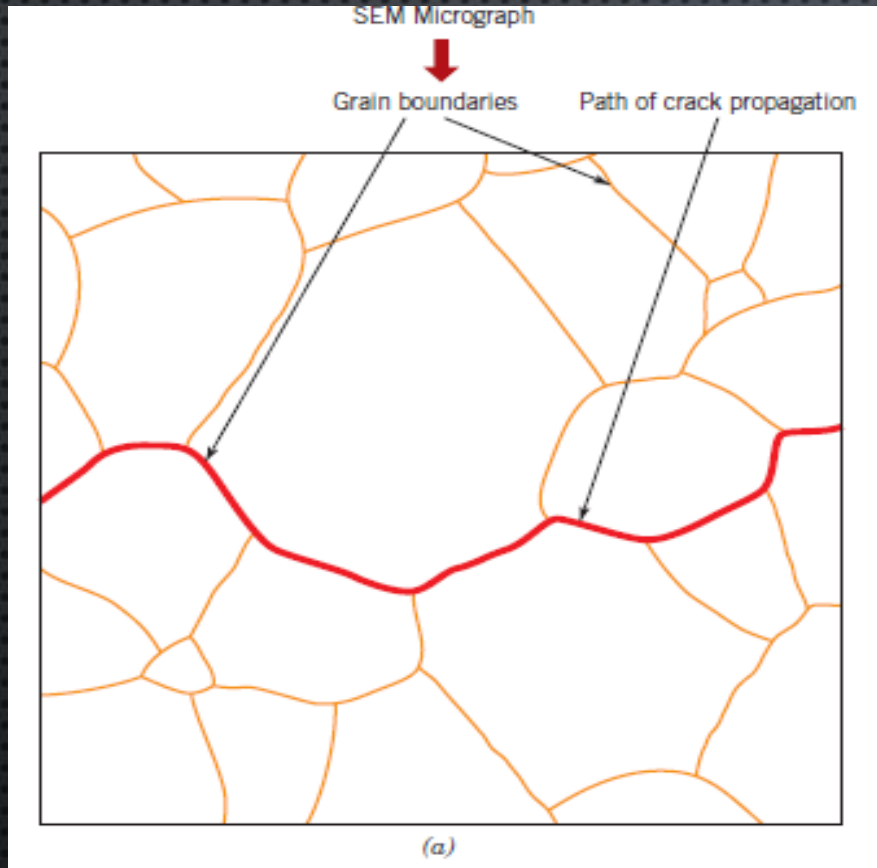
CHARPY TEST



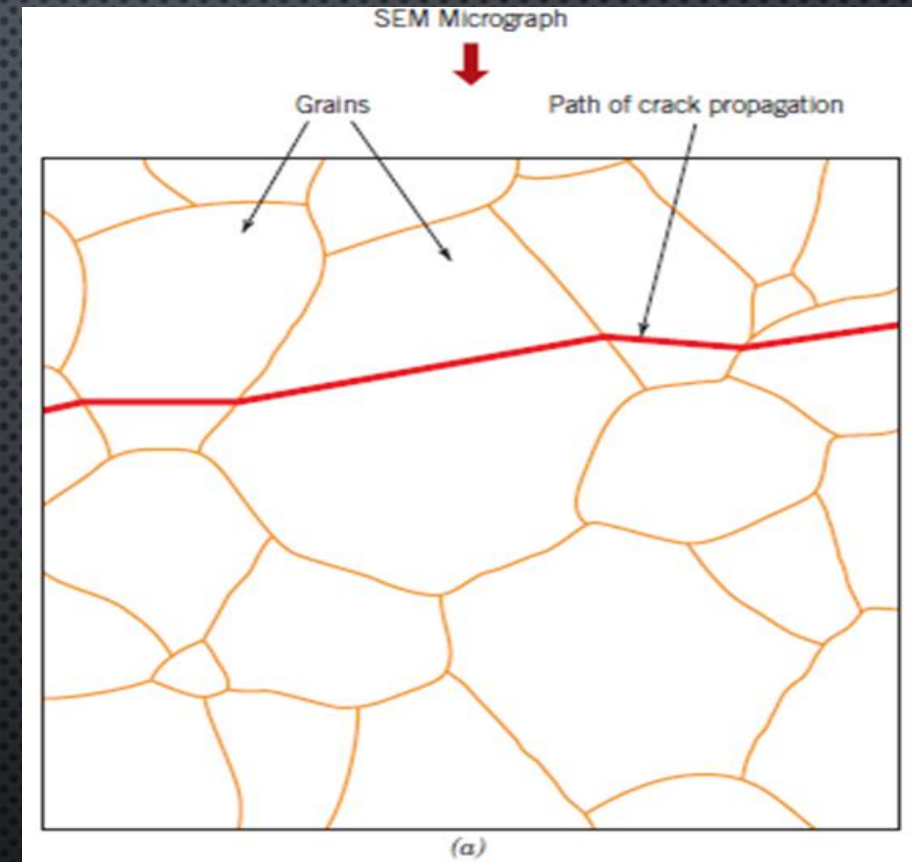
Izod test configuration

IZOD TEST

Different types of fractures include Intergranular fracture (Ductile) and Transgranular fracture (Brittle).



Intergranular (Ductile)



Transgranular (Brittle)

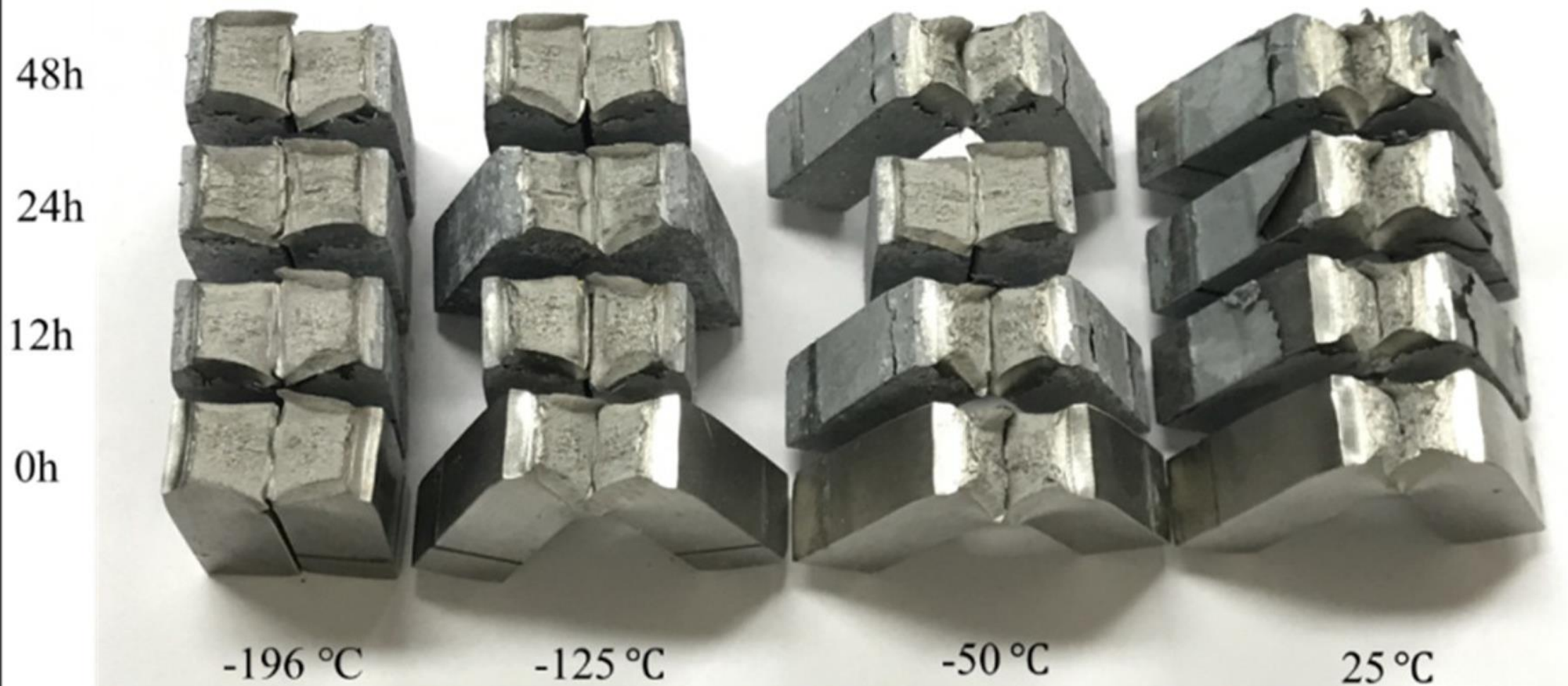
Impact test and Mechanical Properties

- Impact testing assesses a material's toughness, indicating its ability to endure sudden forces.
- The test measures the energy absorbed by the material before it fractures.
- Results help in evaluating the material's suitability for applications involving dynamic loading conditions.
- Impact test data is crucial in industries such as construction, automotive, and aerospace for ensuring safety and reliability.



IMPACT TESTS: CHARPY

- CHARPY TEST MATERIALS THE CHARPY IMPACT TEST IS COMMONLY USED TO EVALUATE THE TOUGHNESS OF VARIOUS MATERIALS, INCLUDING:
- METALS: FERROUS AND NON-FERROUS METALS, SUCH AS STEEL, ALUMINUM, AND COPPER ALLOYS
- PLASTICS: THERMOPLASTICS AND THERMOSETTING PLASTICS
- COMPOSITES: FIBER-REINFORCED COMPOSITES AND LAMINATED MATERIALS



Temperature of Charpy impact test

Feature	Charpy Test	Izod Test
Specimen position	Horizontal	Vertical
Notch location	Facing away from the pendulum	Facing toward the pendulum
Support type	Both ends supported (like a bridge)	One end clamped , other free (like a cantilever)
Common use	Structural & engineering materials	Plastics and some metals
Energy indication	Measured in Joules (energy absorbed)	Same (Joules), but values may differ





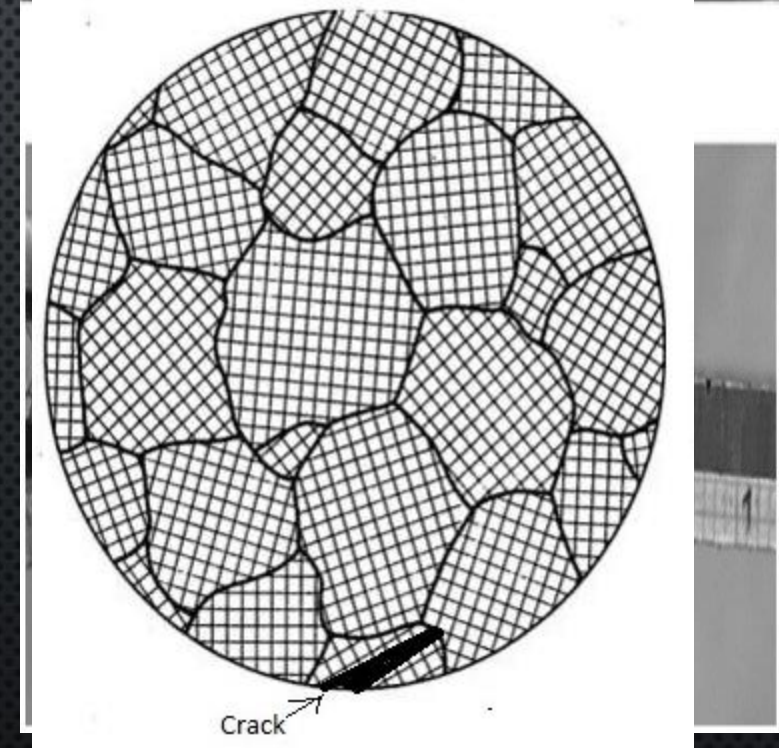
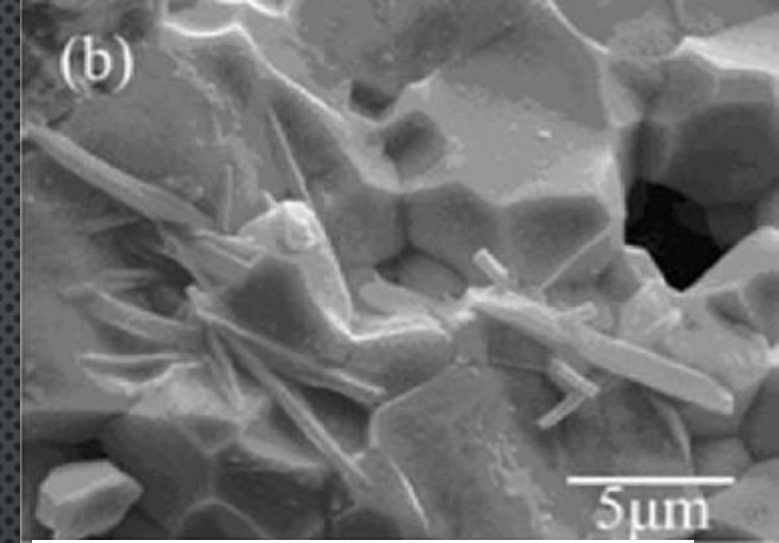
INSTRON

CEAST 10

DO NOT TOUCH THE SPECIMEN

95

- Whisker Bridging
 - Long, thin crystals (whiskers) within a material.
 - Reinforce and span across cracks or voids.
- Fiber Bridging
 - Fibers bridge across cracks or voids.
 - Absorb energy and distribute loads.
- Grain Bridging
 - Grains bridge across crack surfaces.
 - Grain boundaries hinder crack propagation.
- Ductile Second Phase Bridging
 - Ductile phase within a brittle matrix.
 - Deforms plastically and bridges cracks



Mechanism	Type	Function	Common In
Whisker Bridging	Tiny single crystals	Crack bridging, toughening	Ceramics, composites
Fiber Bridging	Long flexible fibers	Resist crack opening, energy absorption	Fiber-reinforced polymers, composites
Grain Bridging	Microstructural grains	Grain holds crack faces	Polycrystalline metals, ceramics
Ductile 2nd Phase Bridging	Soft inclusions/phase	Plastic deformation absorbs energy	Dual-phase steels, composites



Thank you
For
Your attention